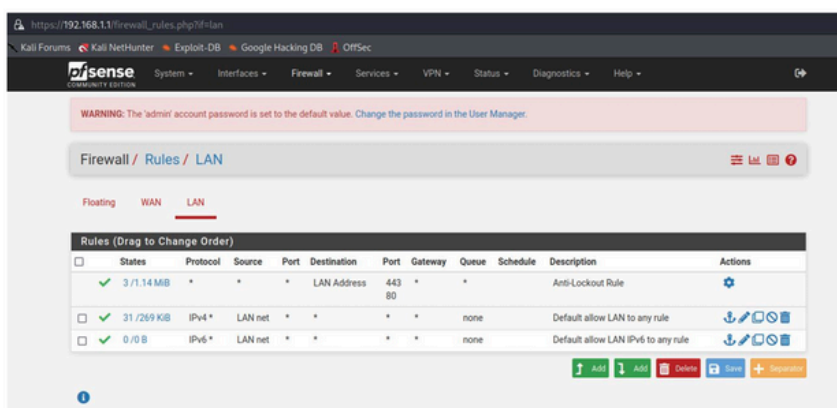


Traccia Esercizio:

Creazione pratica di una regola Firewall.

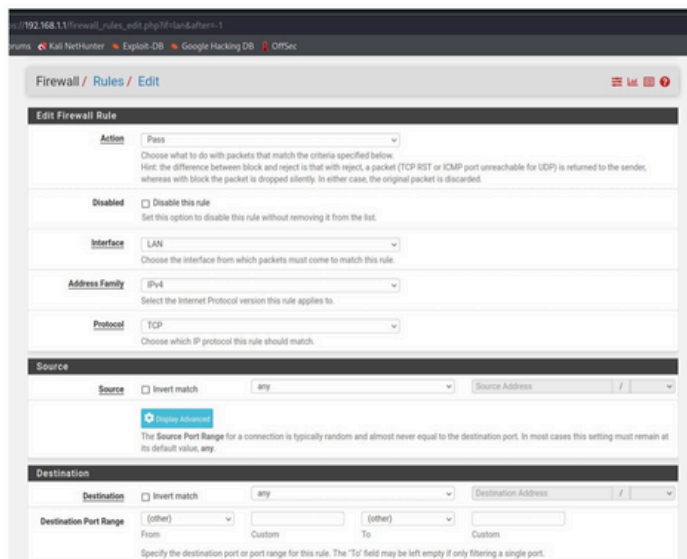
Per la creazione di una regola firewall, andare su Firewall > Rules. In questa sezione si può scegliere su quale interfaccia creare la regola: scegliamo LAN e clicchiamo su ADD (come vedete ci sono 2 add, il primo crea la regola in cima al policy set, la seconda in basso):



Cliccando su add, possiamo aggiungere:

Informazioni generiche:

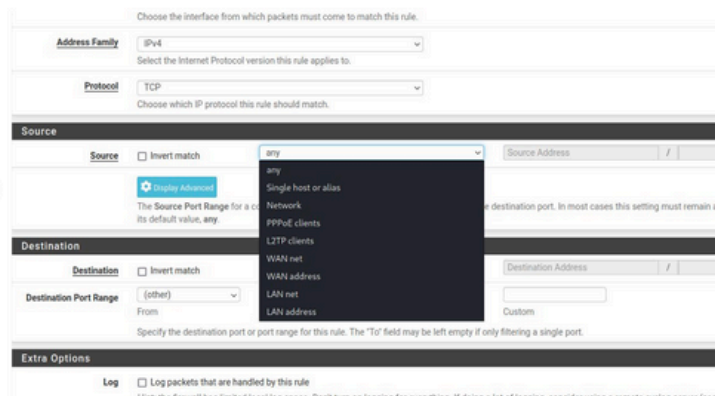
- **Action:** in questa sezione si può scegliere come gestire il traffico analizzato.
- **Interface:** l'interfaccia da dove arrivano i pacchetti.
- **Address family:** Ipv4 oppure ipv6, si sceglie la versione di protocolli ip ai quali applicare la policy.
- **Protocol:** si sceglie il protocollo (e.g., tcp, udp, icmp).



Cliccando su add, possiamo aggiungere:

Informazioni sulla sorgente:

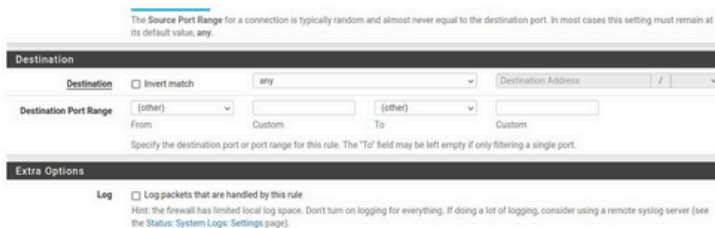
- **Source:** in questa sezione si può scegliere che tipo di sorgente si andrà ad inserire, come un singolo IP, oppure una rete intera. Nel campo valorizzato con «source address» si andranno ad inserire eventualmente gli indirizzi IP o indirizzi rete in notazione CIDR.



Cliccando su add, possiamo aggiungere:

Informazioni sulla destinazione:

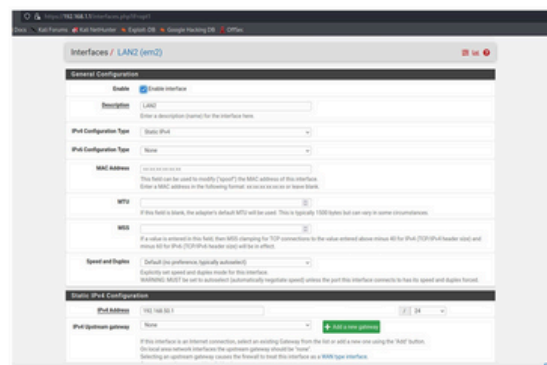
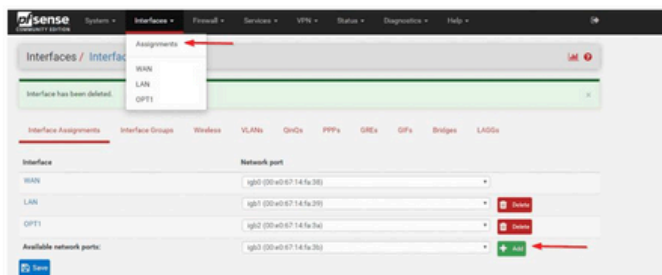
- **Destination:** in questa sezione si può scegliere che tipo di sorgente si andrà ad inserire, come un singolo IP, oppure una rete intera. Nel campo valorizzato con «source address» si andranno ad inserire eventualmente gli indirizzi IP o indirizzi rete in notazione CIDR.
- **Destination port range:** in questa sezione si specificano le porte destinazione. Si possono specificare: singole porte, intervalli, aliases (oggetti di porte custom).



6

Sulla base di quanto visto, creare una regola firewall che **blocchi** l'accesso alla DVWA (su metasploitable) dalla macchina Kali Linux e ne impedisca di conseguenza lo scan. Un requisito fondamentale dell'esercizio è che le macchine Kali e Metasploitable siano su reti diverse, potete aggiungere una nuova interfaccia di rete a Pfsense in modo tale da gestire una ulteriore rete.

Connettetevi poi in Web Gui per attivare la nuova interfaccia e configurarla.



Configurazione di rete:

Kali Linux - Rete interna **intnet** - IP **192.168.50.100**

```
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:1c:a4:6c brd ff:ff:ff:ff:ff:ff
    inet 192.168.50.100/24 brd 192.168.50.255 scope global noprefixroute eth0
        valid_lft forever preferred_lft forever
```

Metasploitable - Rete interna **meta** - IP **192.168.60.150**

```
eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 08:00:27:2b:75:65 brd ff:ff:ff:ff:ff:ff
    inet 192.168.60.150/24 brd 192.168.60.255 scope global eth0
    inet6 fe80::a00:27ff:fe2b:7565/64 scope link
        valid_lft forever preferred_lft forever
```

PFsense

WAN - Nat

LAN - Rete interna **intnet** - 192.168.50.1

META Rete interna **meta** - 192.168.60.1

```
VirtualBox Virtual Machine - Netgate Device ID: b0e7ae32c118177165e3
*** Welcome to pfSense 2.7.2-RELEASE (amd64) on pfSense ***

WAN (wan)      -> vtnet0      -> v4/DHCP4: 10.0.2.15/24
LAN (lan)      -> vtnet1      -> v4: 192.168.50.1/24
META (opt1)    -> vtnet2      -> v4: 192.168.60.1/24

0) Logout (SSH only)          9) pfTop
1) Assign Interfaces          10) Filter Logs
2) Set interface(s) IP address 11) Restart webConfigurator
3) Reset webConfigurator password 12) PHP shell + pfSense tools
4) Reset to factory defaults    13) Update from console
5) Reboot system              14) Enable Secure Shell (sshd)
6) Halt system                15) Restore recent configuration
7) Ping host                  16) Restart PHP-FPM
8) Shell
```

Esecuzione:

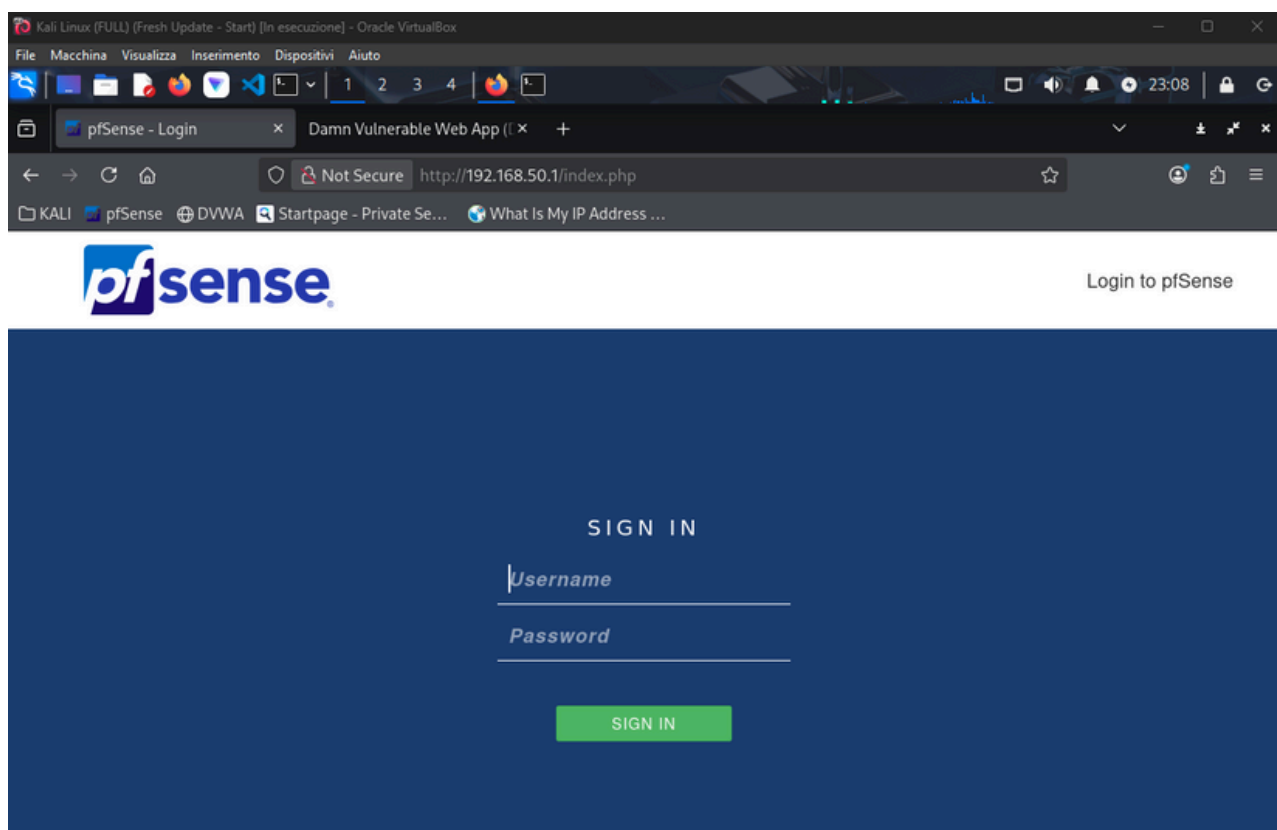
Dalla **Kali Linux** raggiungiamo l'**interfaccia web** di configurazione della **PFsense**

Dal **brower** digitiamo indirizzo del **gateway** 192.168.50.1

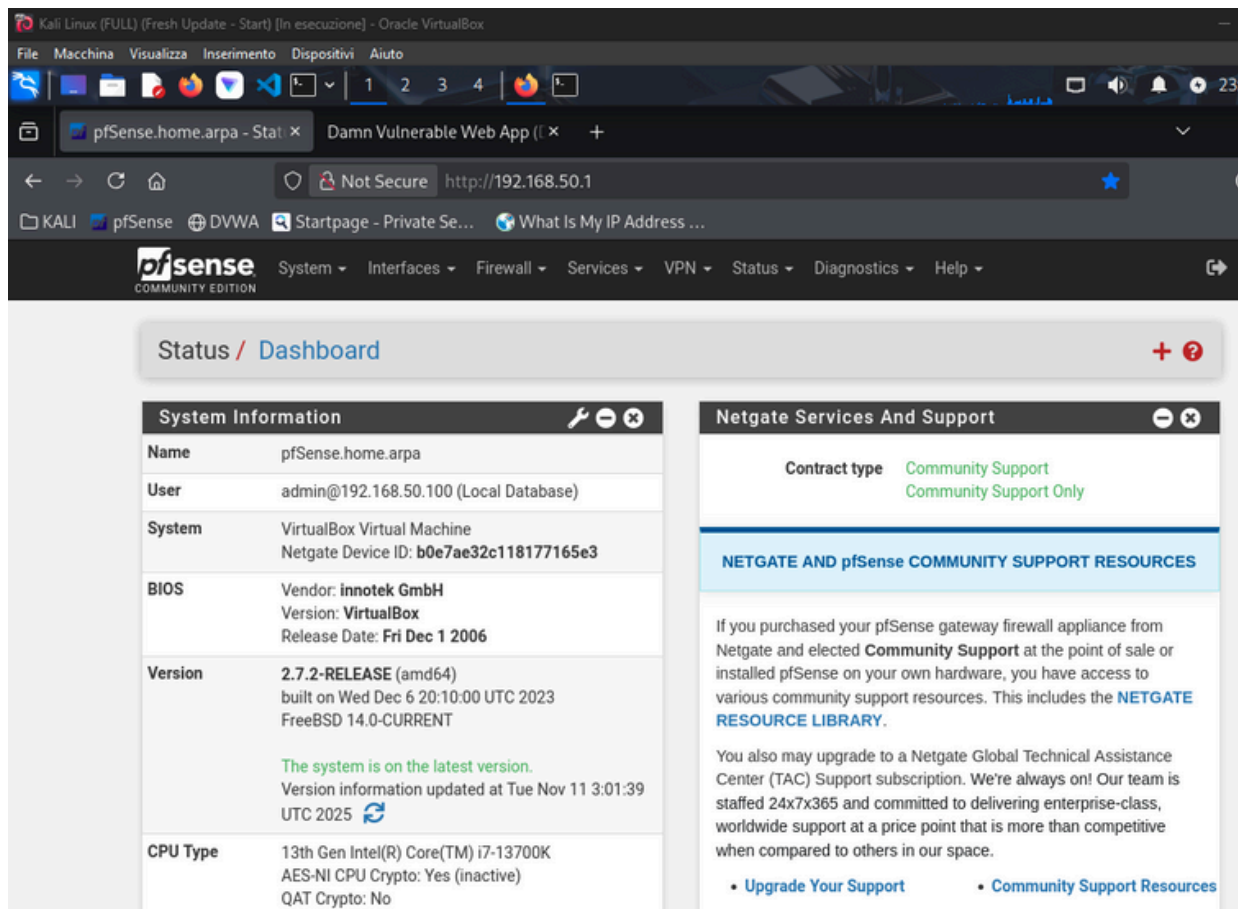
Alla richiesta di login utilizziamo le credenziali di default o quelle configurate.

USERNAME default: admin

PASSWORD default: password

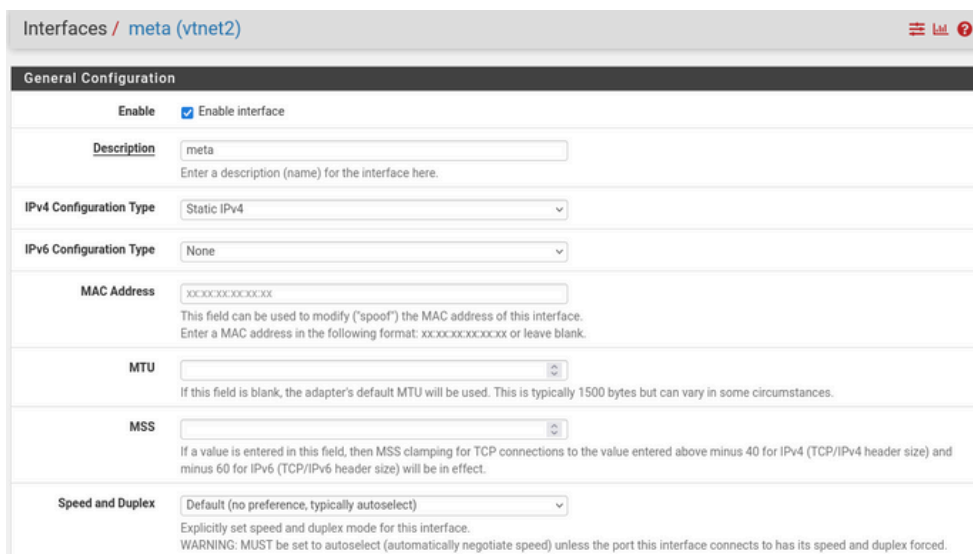


Selezioniamo nel menu grigio in alto **Interfaces, Assignment**



The screenshot shows the pfSense Community Edition dashboard. The top navigation bar includes links for System, Interfaces, Firewall, Services, VPN, Status, Diagnostics, and Help. The main content area is divided into two columns. The left column, titled 'System Information', displays details about the pfSense instance, including the name (pfSense.home.arpa), user (admin@192.168.50.100), system (VirtualBox Virtual Machine), BIOS (Vendor: innotek GmbH, Version: VirtualBox, Release Date: Fri Dec 1 2006), version (2.7.2-RELEASE (amd64)), and CPU type (13th Gen Intel(R) Core(TM) i7-13700K). The right column, titled 'Netgate Services And Support', shows the contract type (Community Support) and provides links to Netgate and pfSense community support resources.

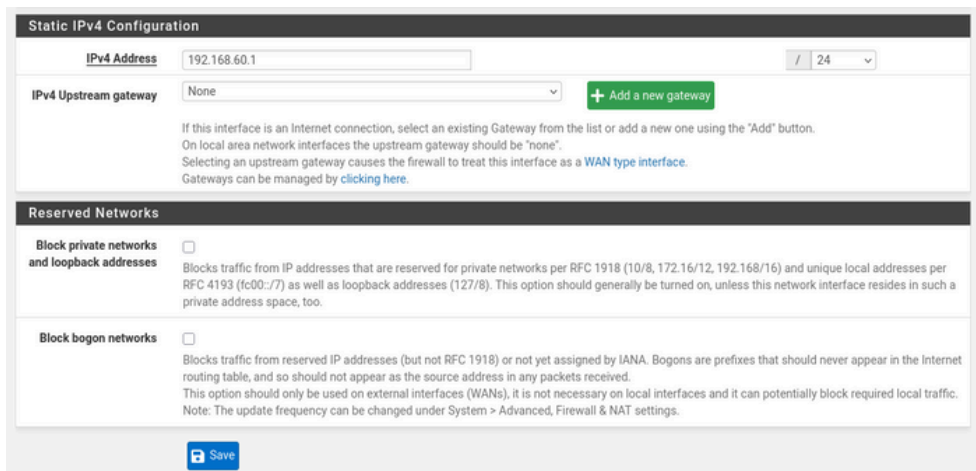
Usiamo **ADD** per aggiungere un **interfaccia**

The screenshot shows the pfSense Interfaces configuration page for the 'meta' interface. The page is titled 'Interfaces / meta (vtnet2)'. The 'General Configuration' section includes the following fields:

- Enable:** A checkbox labeled 'Enable interface' which is checked.
- Description:** A text field containing 'meta'.
- IPv4 Configuration Type:** A dropdown menu set to 'Static IPv4'.
- IPv6 Configuration Type:** A dropdown menu set to 'None'.
- MAC Address:** A text field containing 'xxxxxxxxxx'.
- MTU:** A text field with a spinner, currently set to 1500.
- MSS:** A text field with a spinner, currently set to 1460.
- Speed and Duplex:** A dropdown menu set to 'Default (no preference, typically autoselect)'.

Inseriamo in **IPv4 Address** 192.168.60.1



Static IPv4 Configuration

IPv4 Address 192.168.60.1 / 24

IPv4 Upstream gateway None [+ Add a new gateway](#)

If this interface is an Internet connection, select an existing Gateway from the list or add a new one using the "Add" button.
On local area network interfaces the upstream gateway should be 'none'.
Selecting an upstream gateway causes the firewall to treat this interface as a **WAN type interface**.
Gateways can be managed by [clicking here](#).

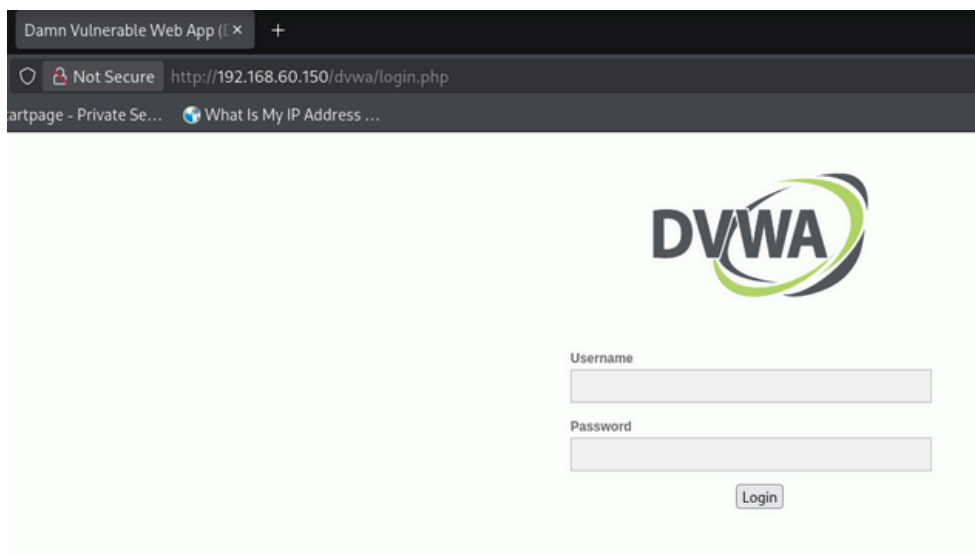
Reserved Networks

Block private networks and loopback addresses ☐
Blocks traffic from IP addresses that are reserved for private networks per RFC 1918 (10/8, 172.16/12, 192.168/16) and unique local addresses per RFC 4193 (fc00::/7) as well as loopback addresses (127/8). This option should generally be turned on, unless this network interface resides in such a private address space, too.

Block bogon networks ☐
Blocks traffic from reserved IP addresses (but not RFC 1918) or not yet assigned by IANA. Bogons are prefixes that should never appear in the Internet routing table, and so should not appear as the source address in any packets received.
This option should only be used on external interfaces (WANs), it is not necessary on local interfaces and it can potentially block required local traffic.
Note: The update frequency can be changed under System > Advanced, Firewall & NAT settings.

[Save](#)

Per accedere alla **DVWA**, dal **browser** della Kali digitiamo indirizzo della Metasploitable
http://192.168.60.150/dvwa/login.php



Damn Vulnerable Web App (x) +

Not Secure http://192.168.60.150/dvwa/login.php

artpage - Private Se... What Is My IP Address ...

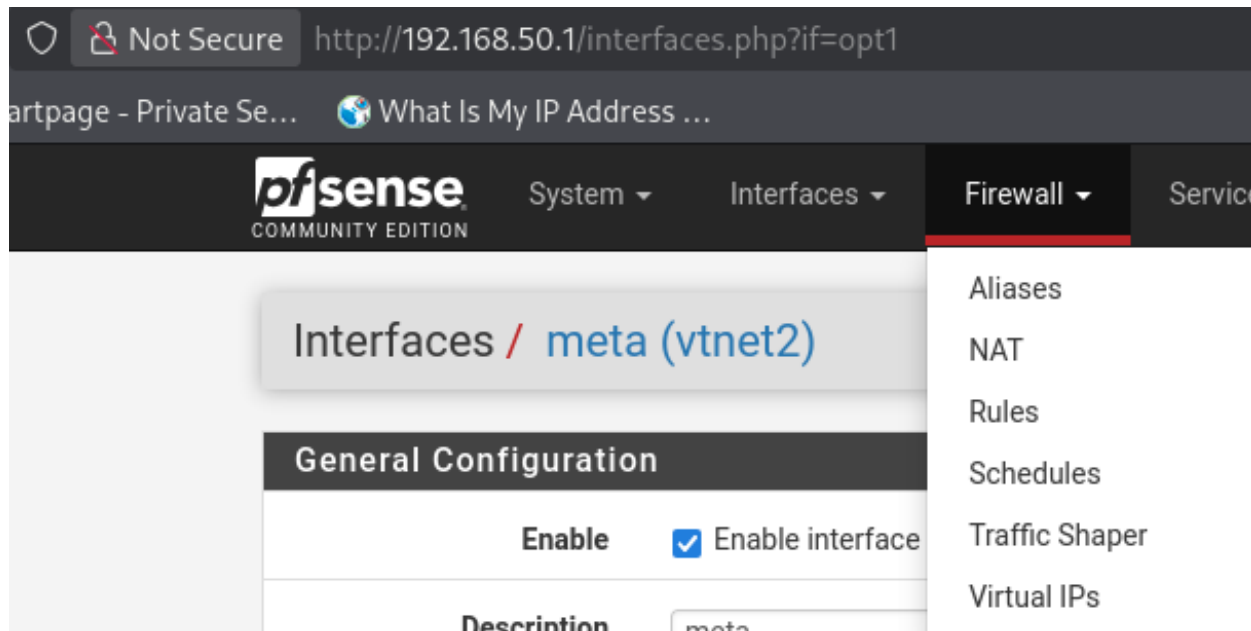
DVWA

Username

Password

[Login](#)

Per **bloccare** l'accesso alla **DVWA** dalla Kali Linux creiamo una nuova regola dal menu in alto **Firewall, Rules**



selezioniamo l'interfaccia di rete **LAN** e poi **ADD**

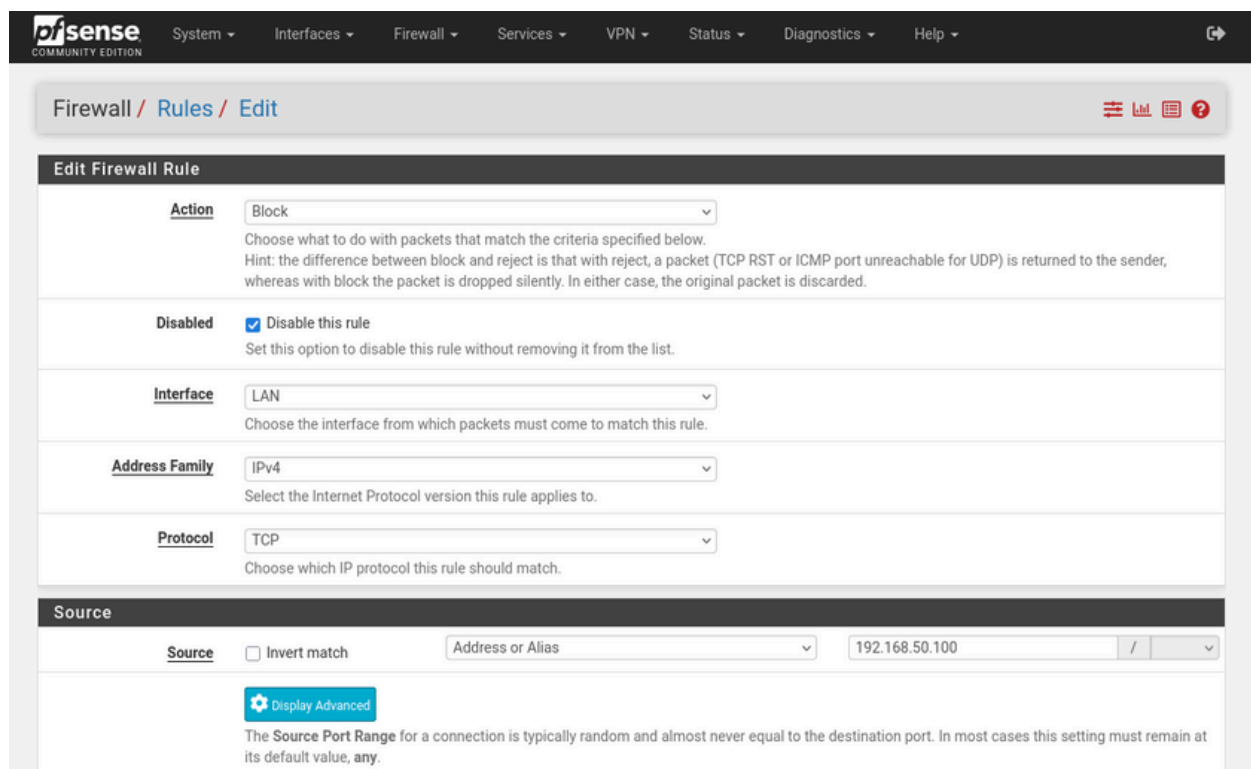
Action: **BLOCK**

Interface: **LAN**

Address Family: **IPv4**

Protocol: **TCP**

Source: **Address or Alias 192.168.50.100**



Destination: **Address or Alias 192.168.60.150**

Destination Port Range: **HTTP 80 to HTTP 80**

Destination

Destination

☐ Invert match

Address or Alias

192.168.60.150

Destination Port Range

HTTP (80)

From

Custom

HTTP (80)

To

Custom

Specify the destination port or port range for this rule. The "To" field may be left empty if only filtering a single port.

Extra Options

Log

☐ Log packets that are handled by this rule

Hint: the firewall has limited local log space. Don't turn on logging for everything. If doing a lot of logging, consider using a remote syslog server (see the [Status: System Logs: Settings](#) page).

Description

A description may be entered here for administrative reference. A maximum of 52 characters will be used in the ruleset and displayed in the firewall log.

Advanced Options

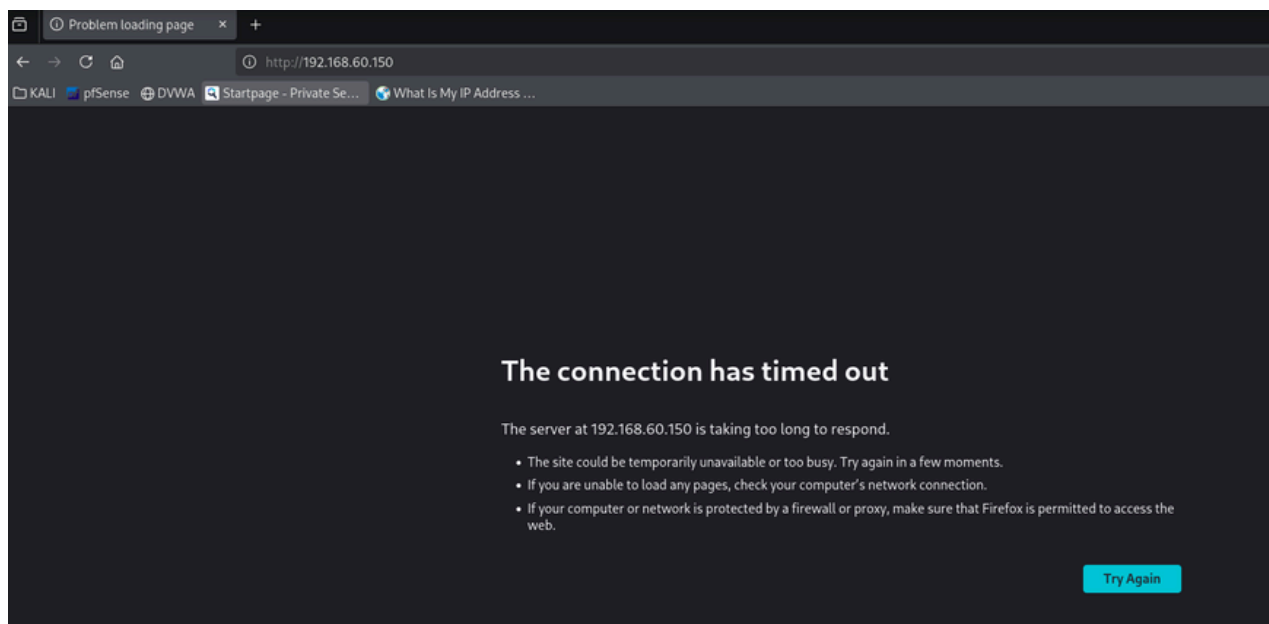
[Display Advanced](#)

Rule Information

Tracking ID	1762836361
Created	11/11/25 04:46:01 by admin@192.168.50.100 (Local Database)
Updated	11/11/25 04:50:58 by admin@192.168.50.100 (Local Database)

[Save](#)

Una volta **applicata la regola** la DVWA **non sara' piu' raggiungibile** dalla Kali Linux



nmap -F 192.168.60.150 ci permette di osservare lo stato di porte e servizi

PORTA 80/TCP FILTERED SERVICE HTTP

```
(marco@vbox)-[~]
$ nmap -F 192.168.60.150
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-11 00:07 EST
Nmap scan report for 192.168.60.150
Host is up (0.0011s latency).
Not shown: 81 closed tcp ports (reset)
PORT      STATE      SERVICE
21/tcp    open      ftp
22/tcp    open      ssh
23/tcp    open      telnet
25/tcp    open      smtp
27/tcp    open      domain
80/tcp    filtered  http
111/tcp   open      rsh
139/tcp   open      netbios-ssn
445/tcp   open      microsoft-ds
513/tcp   open      login
514/tcp   open      shell
2049/tcp  open      nfs
2121/tcp  open      ccproxy-ftp
3306/tcp  open      mysql
5432/tcp  open      postgresql
5900/tcp  open      vnc
6000/tcp  open      X11
8009/tcp  open      ajp13
8080/tcp  open      http-proxy

Nmap done: 1 IP address (1 host up) scanned in 1.39 seconds
```

Conclusioni:

Abbiamo configurato una **regola firewall** tramite **pfSense** per verificare la nostra capacità di utilizzo della piattaforma e **bloccare l'accesso** di un determinato **client** ai contenuti (in questo caso una porta specifica) di **un'altra rete** LAN. L'applicazione di questo tipo di regole risulta particolarmente utile in ambito aziendale, poiché consente di limitare l'accesso dei dipendenti a determinate risorse, **riducendo** così **il rischio** di danni o violazioni.

Inoltre, la possibilità di nascondere lo stato di una porta contribuisce ad aumentarne la sicurezza, proteggendola da potenziali attacchi, sia interni che esterni.

BONUS:

telnet 192.168.60.150 23 stabiliamo una connessione con il protocollo telnet

```
(marco@vbox)-[~]
$ telnet 192.168.60.150 23
Trying 192.168.60.150 ...
Connected to 192.168.60.150.
Escape character is '^]'.
ls
getuid
ls

metasploit

Warning: Never expose this VM to an untrusted network!
Contact: msfdev[at]metasploit.com
Login with msfadmin/msfadmin to get started
```

per bloccare il traffico telnet ora selezioniamo l'interfaccia di rete **LAN** e poi **ADD**

Action: **BLOCK**

Interface: **LAN**

Address Family: **IPv4**

Protocol: **TCP**

Source: **Address or Alias 192.168.50.100**

Destination: **Address or Alias 192.168.60.150**

Destination Port Range: **Telnet (23)**

Edit Firewall Rule

Action

Block

Choose what to do with packets that match the criteria specified below.
Hint: the difference between block and reject is that with reject, a packet (TCP RST or ICMP port unreachable for UDP) is returned to the sender, whereas with block the packet is dropped silently. In either case, the original packet is discarded.

Disabled

☐ Disable this rule
Set this option to disable this rule without removing it from the list.

Interface

LAN

Choose the interface from which packets must come to match this rule.

Address Family

IPv4

Select the Internet Protocol version this rule applies to.

Protocol

TCP

Choose which IP protocol this rule should match.

Source

Source

☐ Invert match

Address or Alias

192.168.50.100

Display Advanced

The Source Port Range for a connection is typically random and almost never equal to the destination port. In most cases this setting must remain at its default value, any.

Destination

Destination

☐ Invert match

Address or Alias

192.168.60.150

Destination Port Range

Telnet (23)

From

Custom

To

Custom

Specify the destination port or port range for this rule. The "To" field may be left empty if only filtering a single port.

proviamo a collegarci con **telnet** dopo aver applicato la **nuova regola**

```
(marco@vbox)-[~]
$ telnet 192.168.60.150
Trying 192.168.60.150 ...
telnet: Unable to connect to remote host: Connection timed out
```

possiamo osservare

23/tcp FILTERED telnet

```
(marco@vbox)-[~]
$ nmap -F 192.168.60.150
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-11 00:24 EST
Nmap scan report for 192.168.60.150
Host is up (0.00051s latency).
Not shown: 81 closed tcp ports (reset)
PORT      STATE      SERVICE
21/tcp    open      ftp
22/tcp    open      ssh
23/tcp    filtered  telnet
25/tcp    open      smtp
53/tcp    open      domain
80/tcp    open      http
111/tcp   open      rpcbind
139/tcp   open      netbios-ssn
445/tcp   open      microsoft-ds
513/tcp   open      login
514/tcp   open      shell
2049/tcp  open      nfs
2121/tcp  open      ccproxy-ftp
3306/tcp  open      mysql
5432/tcp  open      postgresql
5900/tcp  open      vnc
6000/tcp  open      X11
8009/tcp  open      ajp13
8080/tcp  open      http-proxy

Nmap done: 1 IP address (1 host up) scanned in 1.38 seconds
```